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BSIT – 303I

10 Sample Python Codes: Lists and Tuples

Here are 10 sample Python programs for laboratory practice, from basic to intermediate level. Each example includes the source code and sample output.

1. Creating and Displaying a List

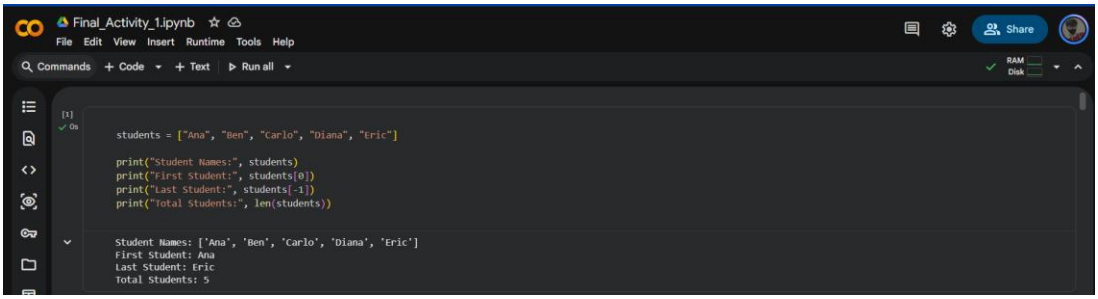
Concept: List creation, indexing, and `len()`.

Python

```
# Sample 1: Student Names

students = ["Ana", "Ben", "Carlo", "Diana", "Eric"]

print("Student Names:", students)
print("First Student:", students[0])
print("Last Student:", students[-1])
print("Total Students:", len(students))
```



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2. Adding and Removing List Items

Concept: `append()`, `insert()`, `remove()`, and `pop()`.

Python

```
# Sample 2: Computer Equipment

equipment = ["Monitor", "Keyboard", "Mouse"]

equipment.append("Printer")
print("After append:", equipment)

equipment.insert(1, "CPU")
print("After insert:", equipment)

equipment.remove("Mouse")
print("After remove:", equipment)

equipment.pop()
print("After pop:", equipment)
```



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3. List Slicing

Concept: Accessing a range of list elements.

Python

```python
# Sample 3: Programming Languages

languages = [
    "Python", "Java", "C++",
    "JavaScript", "PHP", "SQL"
]

print("All Languages:", languages)
print("First Three:", languages[:3])
print("Last Three:", languages[3:])
print("Reversed:", languages[::-1])
```



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4. Sorting and Searching a List

Concept: `sort()`, `in`, `min()`, and `max()`.

Python

```
# Sample 4: Quiz Scores

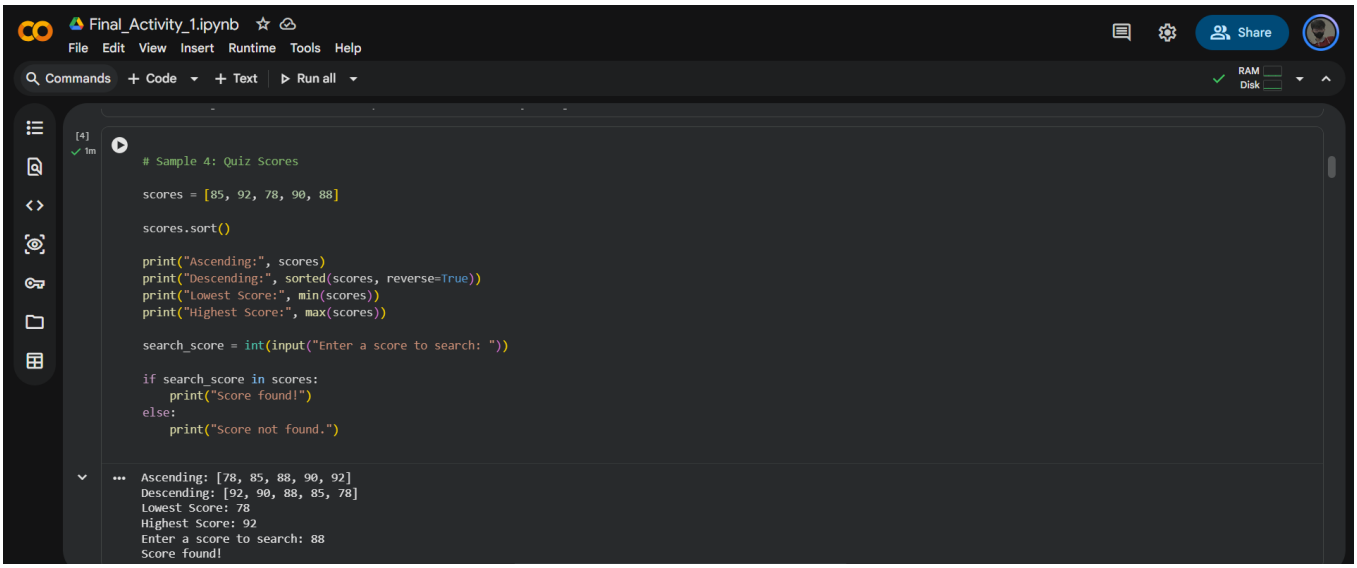
scores = [85, 92, 78, 90, 88]

scores.sort()

print("Ascending:", scores)
print("Descending:", sorted(scores, reverse=True))
print("Lowest Score:", min(scores))
print("Highest Score:", max(scores))

search_score = int(input("Enter a score to search: "))

if search_score in scores:
    print("Score found!")
else:
    print("Score not found.")
```



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5. Computing the Average Using a List

Concept: `sum()`, `len()`, and arithmetic operations.

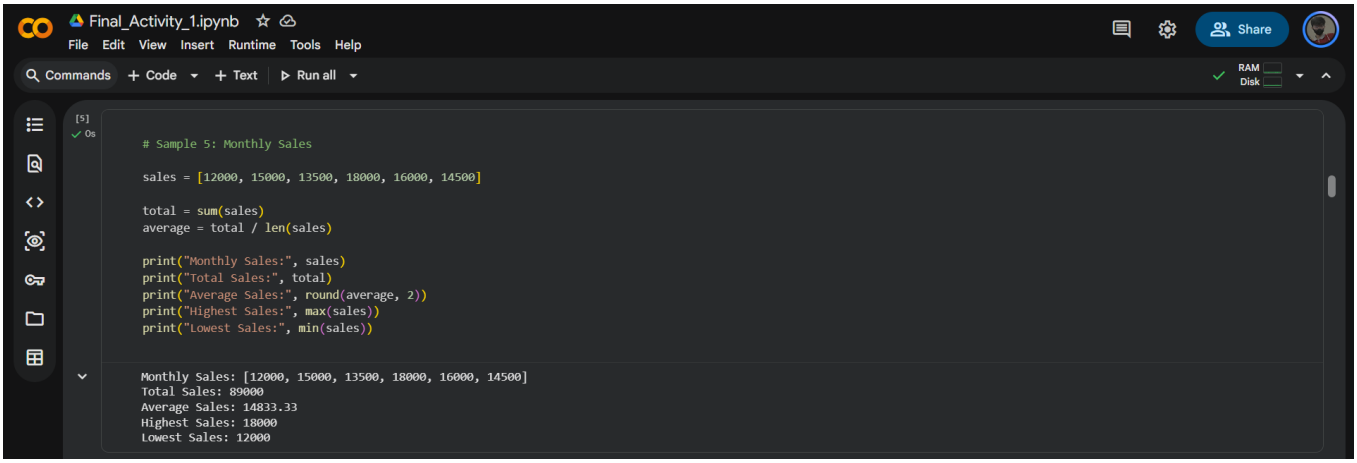
Python

```
# Sample 5: Monthly Sales

sales = [12000, 15000, 13500, 18000, 16000, 14500]

total = sum(sales)
average = total / len(sales)

print("Monthly Sales:", sales)
print("Total Sales:", total)
print("Average Sales:", round(average, 2))
print("Highest Sales:", max(sales))
print("Lowest Sales:", min(sales))
```



6. Creating and Accessing a Tuple

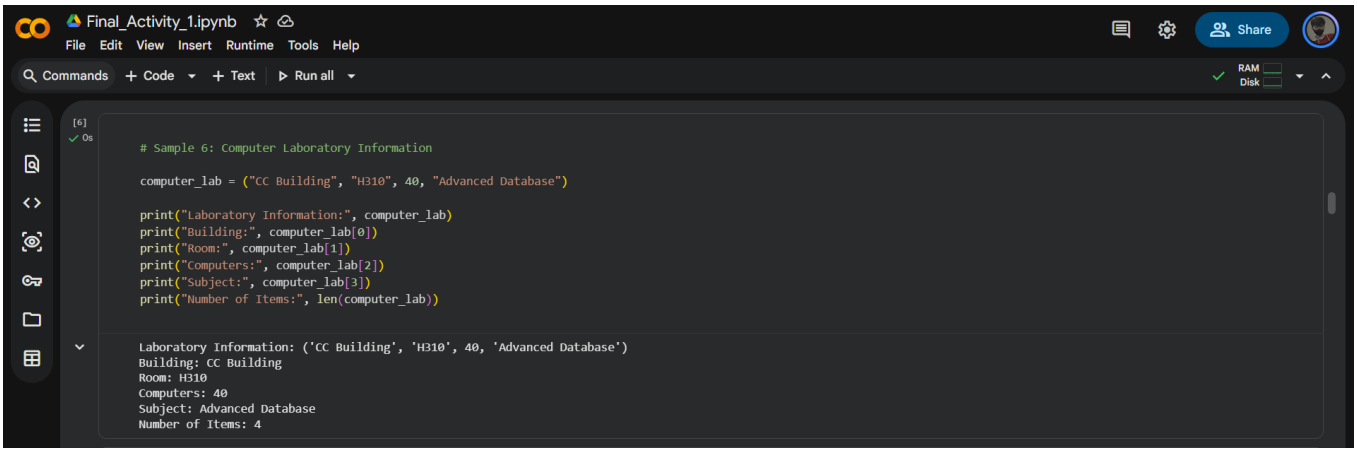
Concept: Tuple creation, indexing, and immutability.

Python

```
# Sample 6: Computer Laboratory Information

computer_lab = ("CC Building", "H310", 40, "Advanced Database")

print("Laboratory Information:", computer_lab)
print("Building:", computer_lab[0])
print("Room:", computer_lab[1])
print("Computers:", computer_lab[2])
print("Subject:", computer_lab[3])
print("Number of Items:", len(computer_lab))
```



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7. Tuple Methods: `count()` and `index()`

Concept: Counting occurrences and finding an element's position.

Python

```
# Sample 7: IT Tools

tools = ("Python", "SQL", "Java", "Python", "Excel", "Python")

print("Tools:", tools)
print("Python Count:", tools.count("Python"))
print("Python First Index:", tools.index("Python"))
print("SQL First Index:", tools.index("SQL"))
```



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8. Combining and Repeating Tuples

Concept: Tuple concatenation (+) and repetition (*).

Python

```
# Sample 8: Course Information

course1 = ("Python", "Database")
course2 = ("Networking", "Cybersecurity")

combined = course1 + course2
repeated = course1 * 2

print("Course 1:", course1)
print("Course 2:", course2)
print("Combined Tuple:", combined)
print("Repeated Tuple:", repeated)
```



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9. Converting a List to a Tuple and Back

Concept: `tuple()` and `list()`.

Python

```
# Sample 9: Subject Enrollment

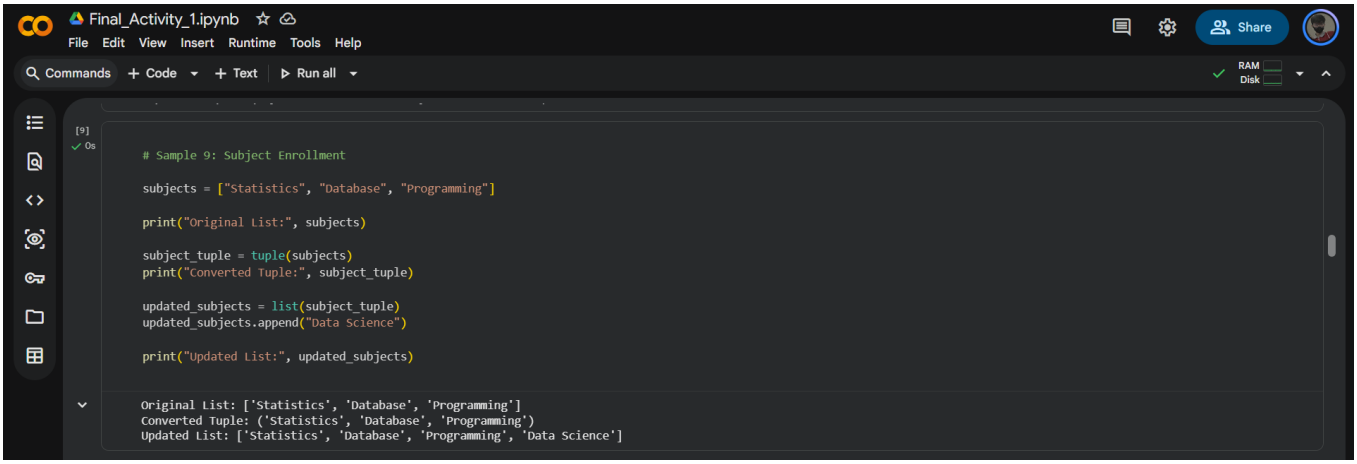
subjects = ["Statistics", "Database", "Programming"]

print("Original List:", subjects)

subject_tuple = tuple(subjects)
print("Converted Tuple:", subject_tuple)

updated_subjects = list(subject_tuple)
updated_subjects.append("Data Science")

print("Updated List:", updated_subjects)
```



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10. Removing Duplicate Items from a List

Concept: Lists, loops, conditional statements, and `append()`.

Python

```
# Sample 10: Remove Duplicate Programming Languages

data = [
    "Python", "Java", "Python",
    "C++", "Java", "SQL", "C++"
]

unique = []

for item in data:
    if item not in unique:
        unique.append(item)

print("Original List:", data)
print("Unique List:", unique)
print("Number of Unique Items:", len(unique))
```



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10 Difficult-Level Python Sample Codes: Lists and Tuples

These examples are suitable for intermediate to advanced laboratory exercises for IT students. They combine lists, tuples, nested data structures, loops, conditional statements, sorting, searching, and data processing.

Sample 1: Remove Duplicates While Preserving Order

Concepts: Lists, loops, membership testing, conditional statements.

Problem: Remove duplicate student names without changing the order of their first appearance.

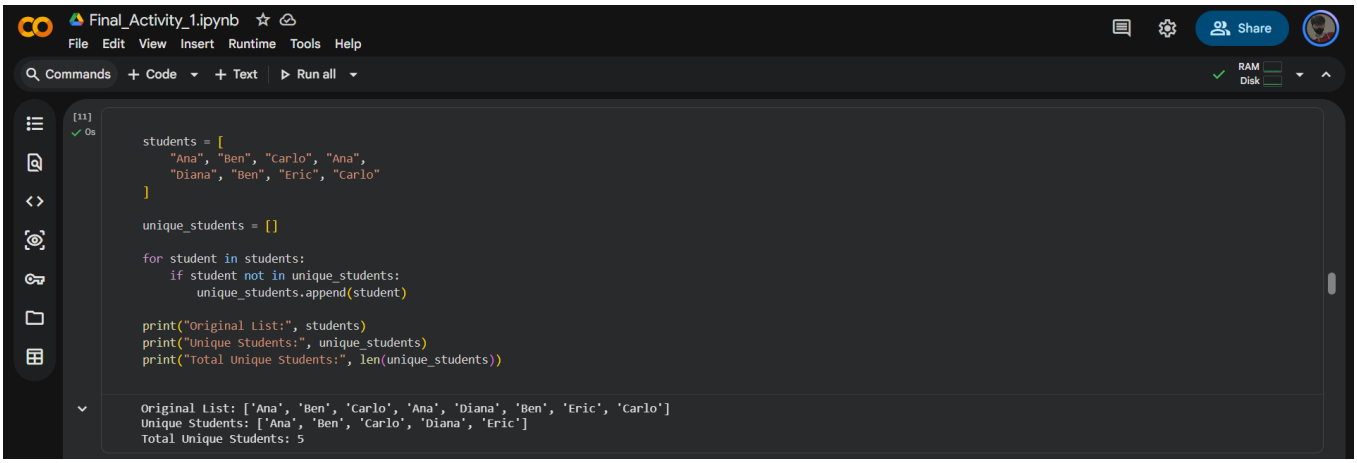
Python

```
students = [
    "Ana", "Ben", "Carlo", "Ana",
    "Diana", "Ben", "Eric", "Carlo"
]

unique_students = []

for student in students:
    if student not in unique_students:
        unique_students.append(student)

print("Original List:", students)
print("Unique Students:", unique_students)
print("Total Unique Students:", len(unique_students))
```



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Sample 2: Find the Second-Highest Unique Score

Concepts: Lists, sets, sorting, duplicate handling.

Problem: Find the second-highest distinct quiz score, even when the highest score appears more than once.

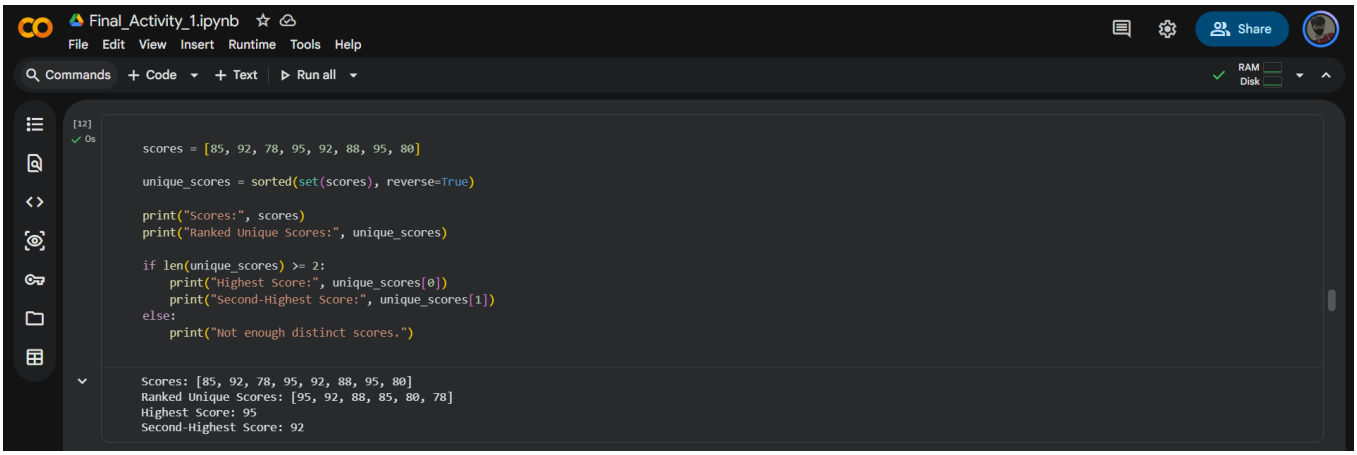
Python

```
scores = [85, 92, 78, 95, 92, 88, 95, 80]

unique_scores = sorted(set(scores), reverse=True)

print("Scores:", scores)
print("Ranked Unique Scores:", unique_scores)

if len(unique_scores) >= 2:
    print("Highest Score:", unique_scores[0])
    print("Second-Highest Score:", unique_scores[1])
else:
    print("Not enough distinct scores.")
```



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Sample 3: Frequency Distribution Using a List

Concepts: Lists, tuples, loops, `count()`, conditional logic.

Problem: Count how many times each distinct programming language appears in a survey response list.

Python

```
responses = [
    "Python", "Java", "Python", "C++",
    "Java", "Python", "SQL", "C++",
    "Java", "SQL", "Python"
]

frequency = []

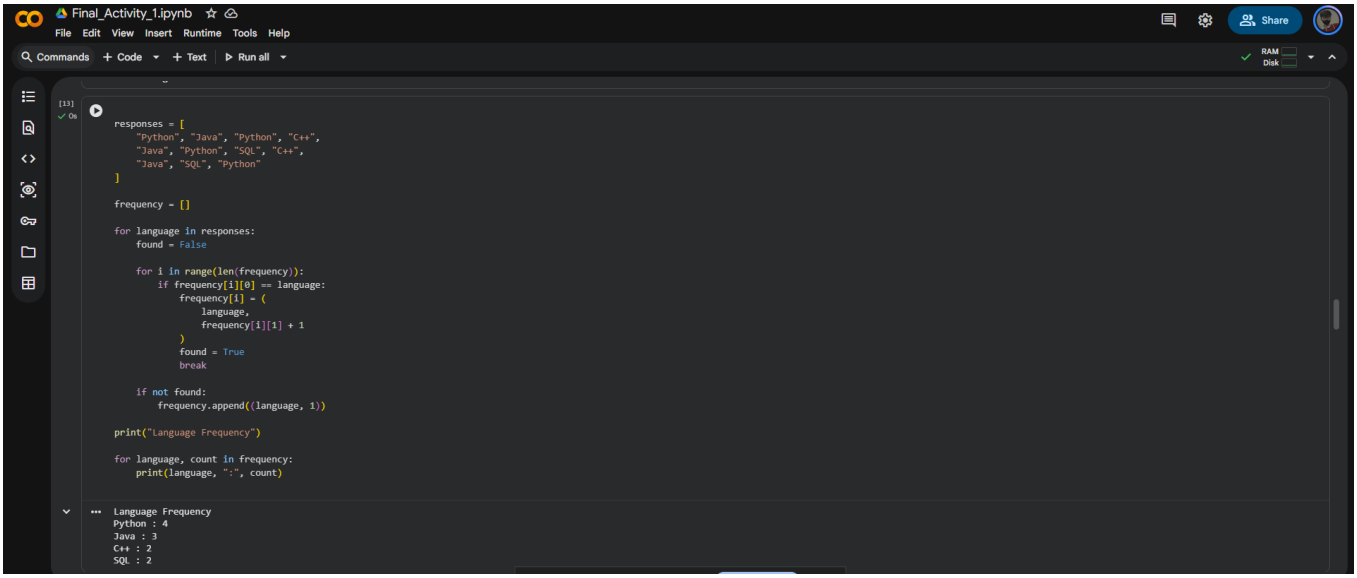
for language in responses:
    found = False

    for i in range(len(frequency)):
        if frequency[i][0] == language:
            frequency[i] = (
                language,
                frequency[i][1] + 1
            )
            found = True
            break

    if not found:
        frequency.append((language, 1))

print("Language Frequency")

for language, count in frequency:
    print(language, ":", count)
```



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Each entry in `frequency` is a tuple containing a language and its count.

Sample 4: Student Ranking System

Concepts: Nested tuples, sorting with `key`, lambda functions.

Problem: Rank students according to their average grades, from highest to lowest.

Python

```python
students = [
    ("Ana", (90, 85, 95)),
    ("Ben", (78, 88, 82)),
    ("Carlo", (92, 91, 89)),
    ("Diana", (85, 87, 90)),
    ("Eric", (92, 91, 89))
]
```

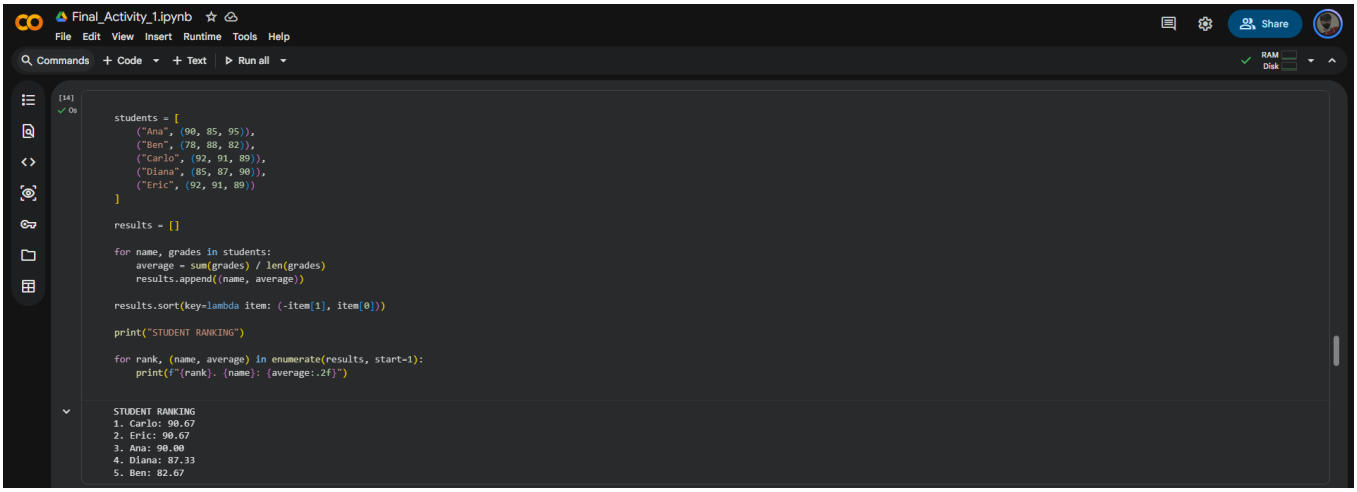
```python
results = []

for name, grades in students:
    average = sum(grades) / len(grades)
    results.append((name, average))

results.sort(key=lambda item: (-item[1], item[0]))

print("STUDENT RANKING")

for rank, (name, average) in enumerate(results, start=1):
    print(f"{rank}. {name}: {average:.2f}")
```



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Sample 5: Matrix Transposition Using Nested Lists

Concepts: Nested lists, indexing, loops, list comprehensions.

Problem: Convert a matrix with three rows and four columns into a matrix with four rows and three columns.

</> Python

```
matrix = [
    [10, 20, 30, 40],
    [50, 60, 70, 80],
    [90, 100, 110, 120]
]

rows = len(matrix)
columns = len(matrix[0])

transposed = []
```

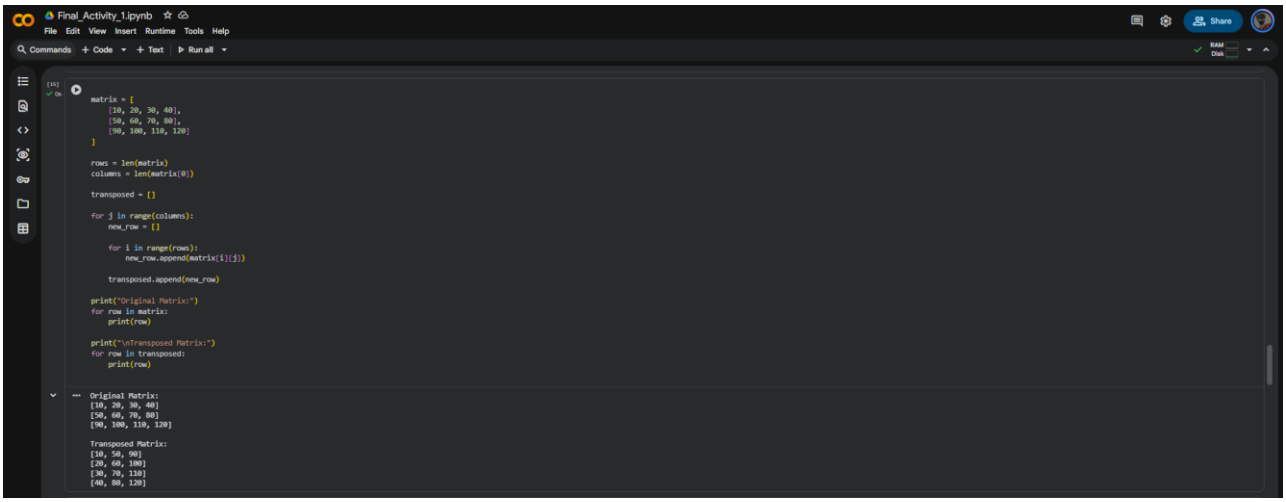
```
for j in range(columns):
    new_row = []

    for i in range(rows):
        new_row.append(matrix[i][j])

    transposed.append(new_row)

print("Original Matrix:")
for row in matrix:
    print(row)

print("\nTransposed Matrix:")
for row in transposed:
    print(row)
```



Sample 6: Common and Unique Elements Between Two Lists

Concepts: Lists, sets, membership testing, set operations.

Problem: Compare two lists of students enrolled in different IT electives.

Python

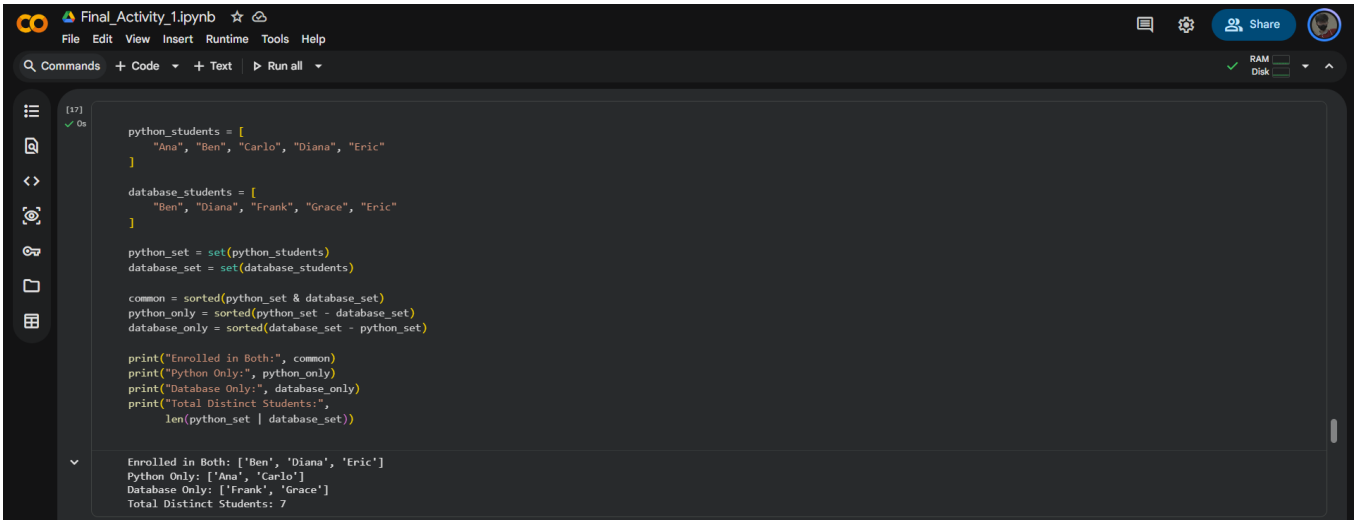
```
python_students = [
    "Ana", "Ben", "Carlo", "Diana", "Eric"
]

database_students = [
    "Ben", "Diana", "Frank", "Grace", "Eric"
]

python_set = set(python_students)
database_set = set(database_students)

common = sorted(python_set & database_set)
python_only = sorted(python_set - database_set)
database_only = sorted(database_set - python_set)
```

```
print("Enrolled in Both:", common)
print("Python Only:", python_only)
print("Database Only:", database_only)
print("Total Distinct Students:",
      len(python_set | database_set))
```



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Sample 7: Rotate a List by a User-Defined Number

Concepts: List slicing, modulo operator, user input.

Problem: Rotate a list of tasks to the right by a number of positions entered by the user.

Python

```
tasks = [
    "Design ERD",
    "Create Tables",
    "Insert Data",
    "Test Queries",
    "Prepare Report"
]

k = int(input("Rotate right by how many positions? "))

if tasks:
    k = k % len(tasks)
    rotated = tasks[-k:] + tasks[:-k] if k else tasks[:]
else:
    rotated = []

print("Original Tasks:", tasks)
print("Rotated Tasks:", rotated)
```



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Sample 8: Merge Two Lists of Records Using Tuples

Concepts: Tuples, dictionaries, loops, record matching.

Problem: Merge student information and grades using student ID as the key.

Python

```
student_info = [
    (101, "Ana"),
    (102, "Ben"),
    (103, "Carlo"),
    (104, "Diana")
]

student_grades = [
    (103, 91),
    (101, 95),
    (104, 88),
    (102, 84)
]

grade_lookup = dict(student_grades)
merged_records = []
```

```
for student_id, name in student_info:
    if student_id in grade_lookup:
        merged_records.append(
            (student_id, name, grade_lookup[student_id])
        )

print("ID | Name | Grade")

for record in merged_records:
    print(record[0], "|", record[1], "|", record[2])
```



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Sample 9: Find Pairs That Equal a Target Sum

Concepts: Nested loops, tuples, list indexing, conditional statements.

Problem: Find all unique pairs of quiz scores whose sum equals a target value.

Python

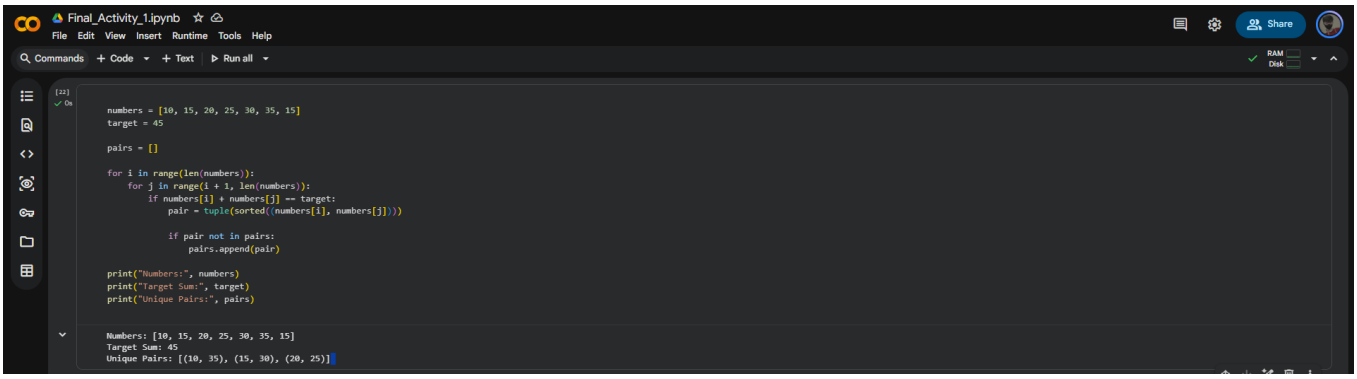
```python
numbers = [10, 15, 20, 25, 30, 35, 15]
target = 45

pairs = []

for i in range(len(numbers)):
    for j in range(i + 1, len(numbers)):
        if numbers[i] + numbers[j] == target:
            pair = tuple(sorted((numbers[i], numbers[j])))

            if pair not in pairs:
                pairs.append(pair)

print("Numbers:", numbers)
print("Target Sum:", target)
print("Unique Pairs:", pairs)
```



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Sample 10: Inventory Management Using Lists and Tuples

Concepts: Lists of tuples, searching, updating records, filtering, sorting.

Problem: Develop a small inventory program that calculates stock values, identifies low-stock products, and ranks products by total inventory value.

Python

```
inventory = [
    ("Mouse", 10, 350),
    ("Keyboard", 5, 650),
    ("Monitor", 3, 5500),
    ("USB Drive", 20, 250),
    ("Headset", 4, 1200)
]

low_stock_limit = 5
total_inventory_value = 0
low_stock = []
stock_values = []
```

```
for name, quantity, price in inventory:
    value = quantity * price
    total_inventory_value += value

    stock_values.append((name, value))

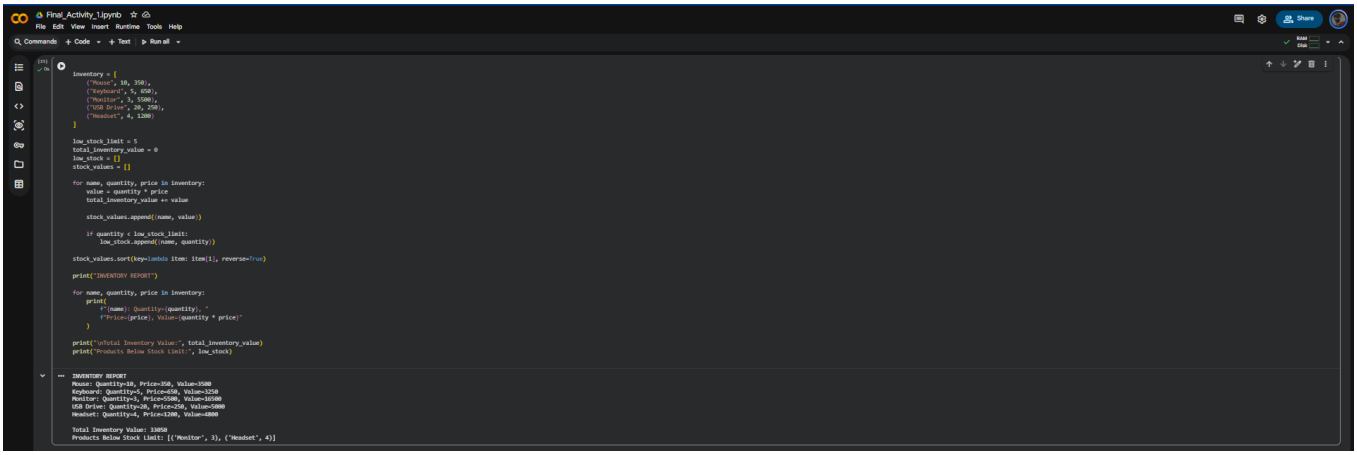
    if quantity < low_stock_limit:
        low_stock.append((name, quantity))

stock_values.sort(key=lambda item: item[1], reverse=True)

print("INVENTORY REPORT")
```

```
for name, quantity, price in inventory:
    print(
        f"{name}: Quantity={quantity}, "
        f"Price={price}, Value={quantity * price}"
    )

print("\nTotal Inventory Value:", total_inventory_value)
print("Products Below Stock Limit:", low_stock)
```



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Summary of Python List and Tuple Operations

Operation	List	Tuple
Create	`items = [1, 2, 3]`	`items = (1, 2, 3)`
Access an item	`items[0]`	`items[0]`
Count elements	`len(items)`	`len(items)`
Count occurrences	`items.count(2)`	`items.count(2)`
Find index	`items.index(2)`	`items.index(2)`
Add elements	`append()`, `insert()`	Not directly supported
Remove elements	`remove()`, `pop()`	Not directly supported